

Eng. mechanics

It is the branch of science that deals with the forces that changes the states of the body.

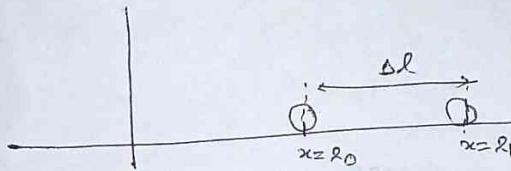
OR

It is the branch of science that deals with the forces and its effects on the bodies.

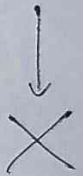
EXTERNAL

INTERNAL

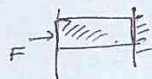
→ Stress + Strain



velocity, displacement



Note:-



Clearly, no external effect but internal effect will be prevalent, these internal effects are stress, strain, etc.

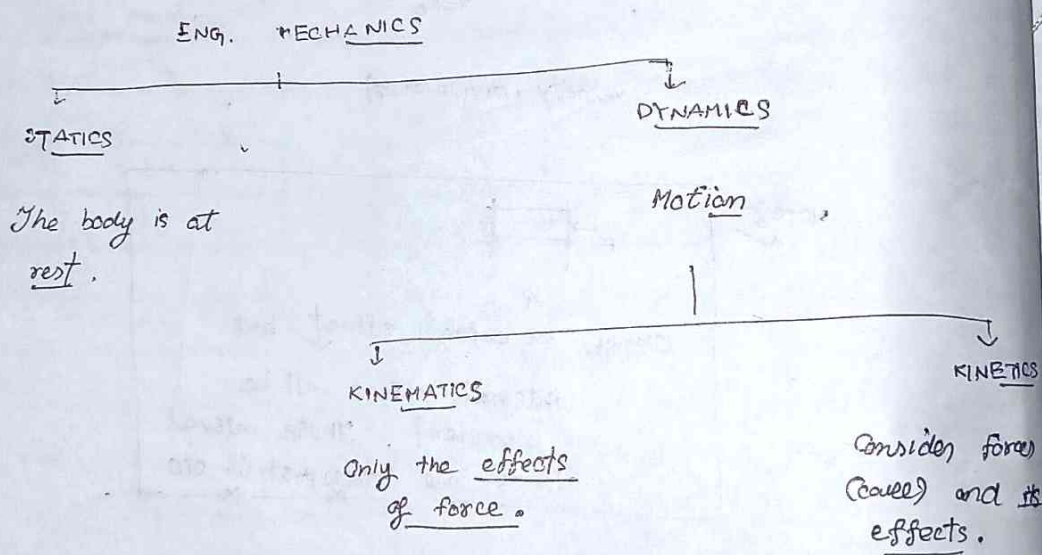
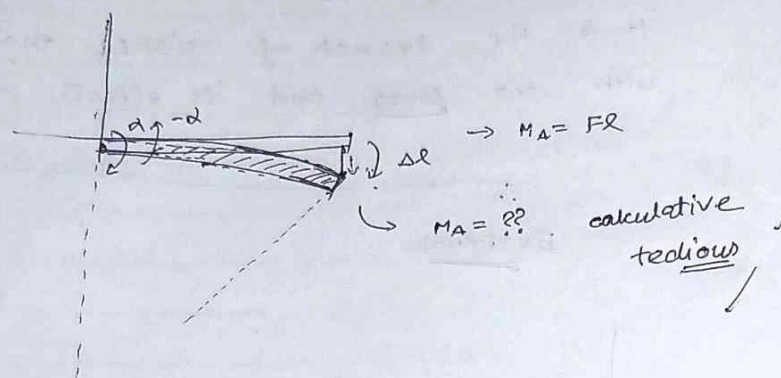
* In engineering mechanics, we'll learn about the

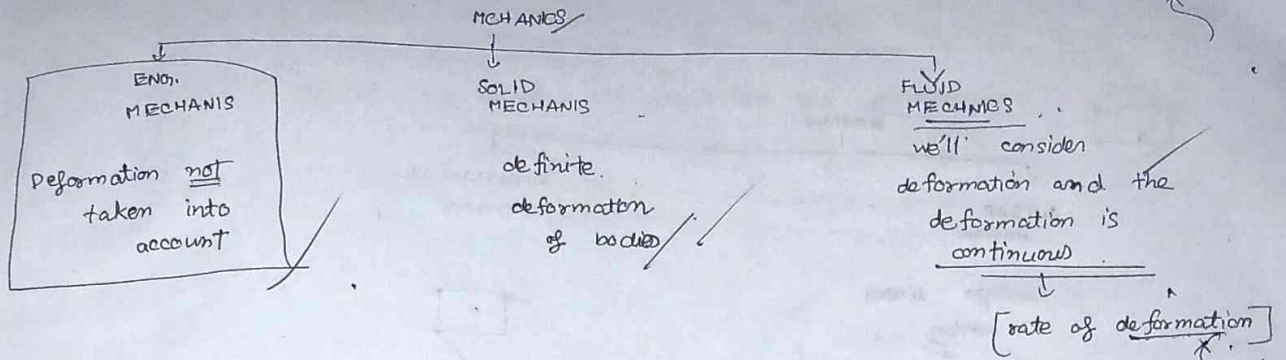
EXTERNAL effect only.

* select only RIGID bodies.

** Rigid body idealisation

After the application of forces, there will not be any change in the relative distance b/w the molecules of the body.



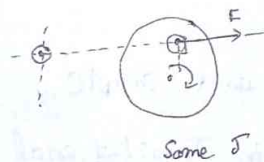
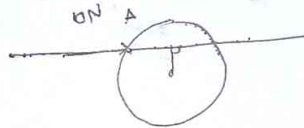


Mech
 Rotat
 Vint

Force : It is the action of one-body into another body that will try to change the state of the motion of that body.

Force is a vector quantity

- ⌘ Magnitude
- ⌘ Line of action
- ⌘ Point of application

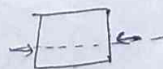


Principle of Transmissibility

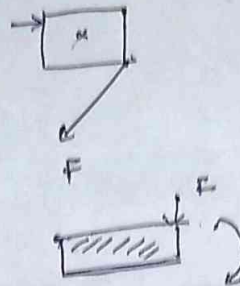
keeping the magnitude of force unaltered, if u change the point of application of force along the line of action there will be no change in the external effect.



EXPANSION



COMPRESSION



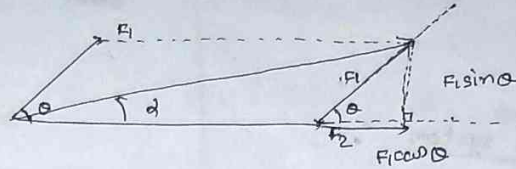
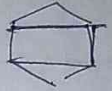
$d \perp F$
 $Fd = M$

The minimum rotation
 by which F meets
 the axis gives the
 direction of Moment
arm

Note :- Moment of Force and Couple
 differs only in Translational
motion

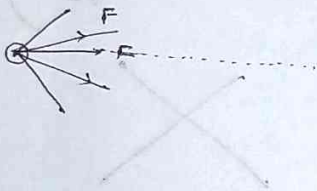
Addition of Forces

Parallelogram: The sides of the parallelogram are parallel



$$F_R^2 = (F_2 + F_1 \cos \theta)^2 + (F_1 \sin \theta)^2$$

$$|F_R| = \sqrt{F_1^2 + F_2^2 + 2F_1 F_2 \cos \theta}$$

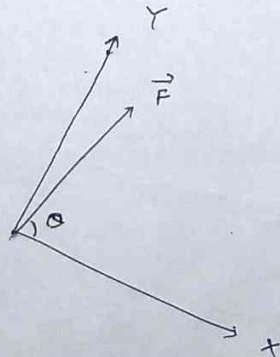


$$\tan \alpha = \frac{F_1 \sin \theta}{F_2 + F_1 \cos \theta}$$

$$\alpha = \tan^{-1} \left[\frac{F_1 \sin \theta}{F_2 + F_1 \cos \theta} \right]$$

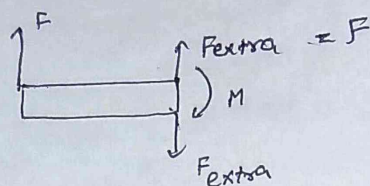
Resolution of Forces

① we'll split the force into 2 perpendicular components



$$x = |F| \cos \theta \quad ; \quad y = F \sin \theta$$

② we'll split the force into one force and one couple.



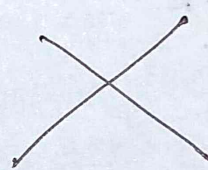
Forcing system :

↓
COPLANER
FORCING SYSTEM ✓

* all forces will be in
same plane
(for 2D-confg.)

↓
NON-COPLANER
FORCING
SYSTEM

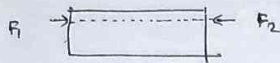
All Forces we be in
different plane
(3D-confg.)



co-planar forcing types

↓
COPLANER and
COLLINEAR
FORCING
SYSTEM

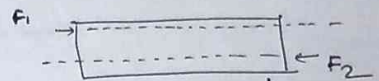
↓
The line of action of the
forces is same



* F_1 & F_2 are COC forces

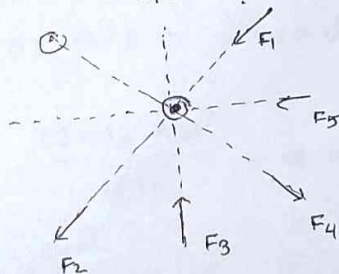
↓
COPLANER and
PARALLEL
FORCING
SYSTEM

↓
The line of action of
forces will be parallel
to each other



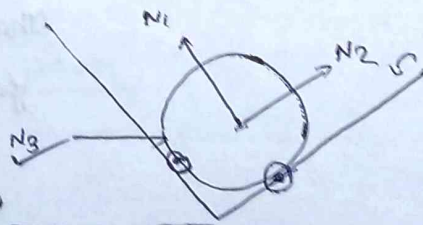
* F_1 & F_2 are COP forces

↓
COPLANER and
CONCURRENT
FORCING
SYSTEM

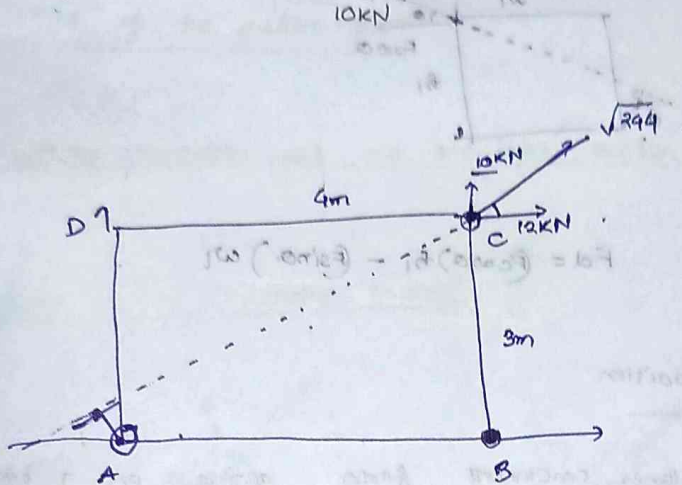
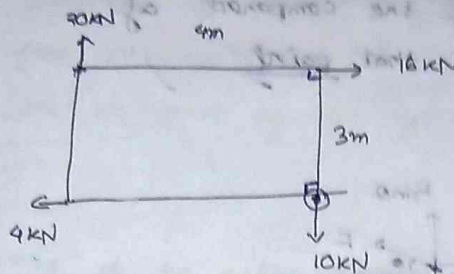


F_1, F_2, F_3, F_4 and F_5 are co-current
forces as their line of actions
meet at the same point

NON-PARALLEL
and non-concurrent
force system



Q. Find the direction and magnitude of following force



$$\frac{144}{100} = \frac{244}{244}$$

$$|\vec{r}| = \sqrt{244}$$

$$\theta = \tan^{-1}\left(\frac{5}{6}\right)$$

with respect to point A.

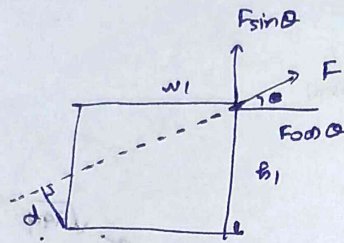
$$0 + 0 + 10 \times 4 + 10 \times 3 = \sqrt{244} \times d$$

$$\frac{48 + 40 \times 10^3}{\sqrt{244}} = d$$

$$\frac{88 \times 10^3}{\sqrt{244}} = d$$

VARIGNON'S LAW

The moment of a force about a point is equal to the sum of the moment of the component of that force ~~of that point~~ about that point. ✓



$$Fd = (F \cos \theta) b_1 - (F \sin \theta) w_1$$

Polygonal law of vector addition

If a number of coplanar, concurrent forces acting on a body are represented by sides of a polygon, taken in order, their resultant force is represented by the closing side taken in opposite order.

Triangle of forces rule

$$\vec{b} \times \vec{a} + \vec{a} \times \vec{c} + \vec{c} \times \vec{b} = 0$$

FBD: It is the sketch of a selected system consisting of one body or a part of a body or some collection of interconnecting bodies completely isolated from other parts of the system showing the interconnection with all parts of the system through force.

Nature of the reaction forces

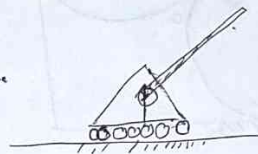
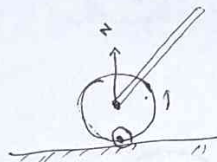
Types of supports and the associated reaction forces

① Roller support / Frictionless support

$$\sin \alpha - \sin \beta$$

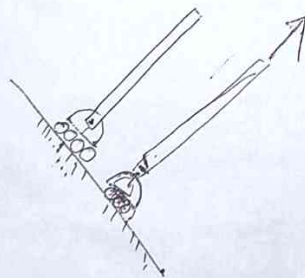
$$2 \sin \left(\frac{\alpha - \beta}{2} \right) \cos \left(\frac{\alpha + \beta}{2} \right) = \sin \beta$$

$$\sqrt{2} \sin \left(\frac{\alpha - \beta}{2} \right) = \sin \beta$$

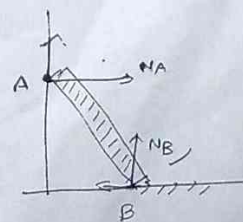
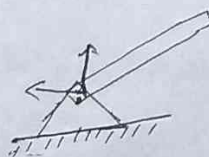


→ when the support resists linear motion, the reaction will be one one force.

→ when the support resists rotational motion, reaction will be moment.



② Hinged support

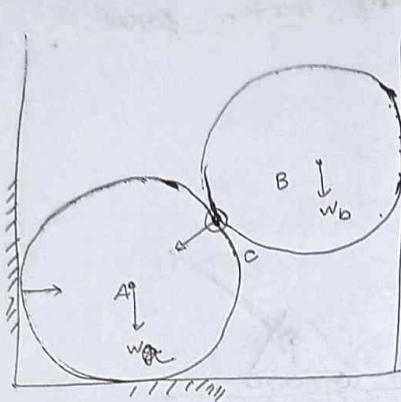


Fixed support

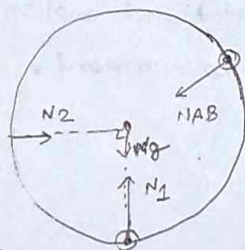


Fixed support

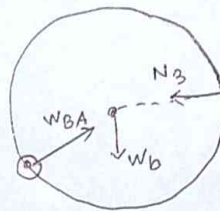
Q/



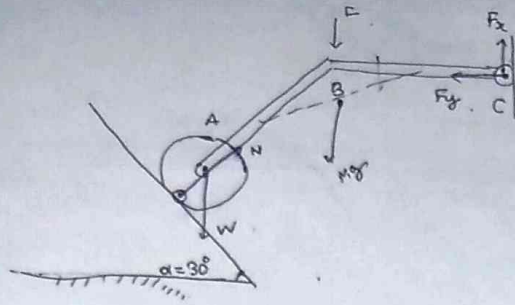
FBD of WA



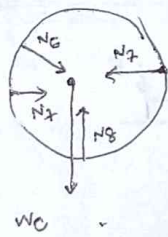
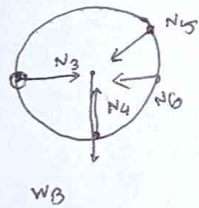
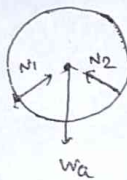
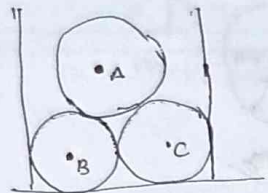
FBD of B



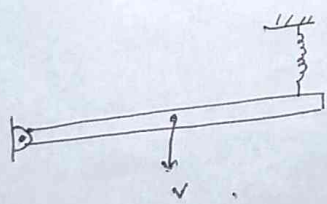
Problem-1



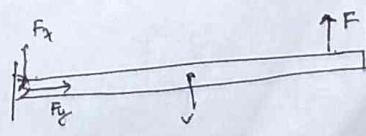
Problem-2



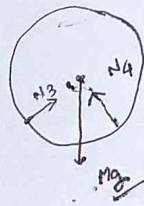
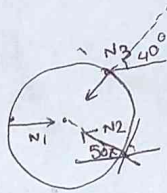
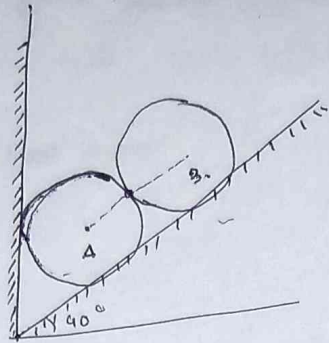
Problem-3



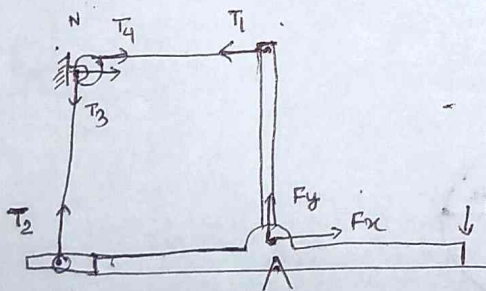
FBD



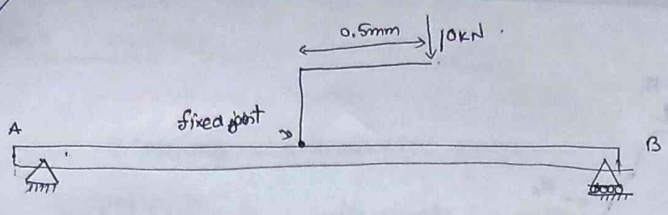
Problem - 4



Problem - 5

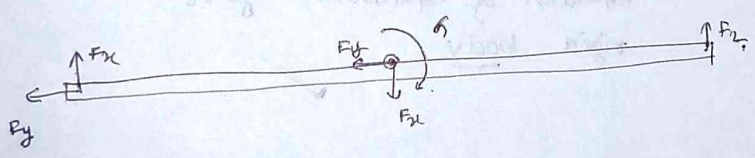
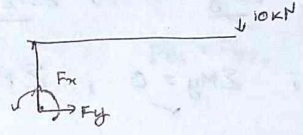
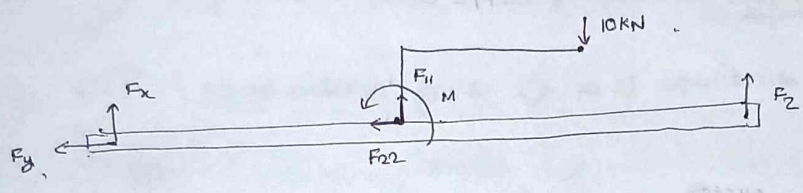


Q 4/W

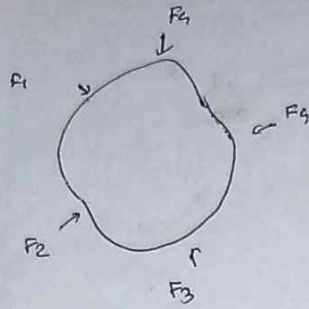


Shivam
9900119356

Solu n^o



Equilibrium of Forcing systems



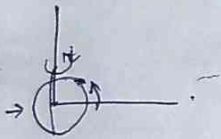
$$\sum \vec{F} = 0$$

$$\sum M = 0$$

For a 3D - system

$$\left[\begin{array}{l} \sum F_x = 0 ; \sum F_y = 0 ; \sum F_z = 0 \\ \sum M_x = 0 ; \sum M_y = 0 ; \sum M_z = 0 \end{array} \right]$$

These 6 eq's are called the
equation of equilibrium of a
rigid body.



$$\rightarrow \sum F_x = \sum F_y = 0$$

$$\sum M_z = 0$$

→ There are three equilibrium
equations of equilibrium of
a 2D - rigid body

→ We'll use these eq's to
calculate the unknown forces
acting on the body.

Based on the no. of unknown forces and no of equilibrium equations available.

① Statically determinate problem

② Statically indeterminate problem.

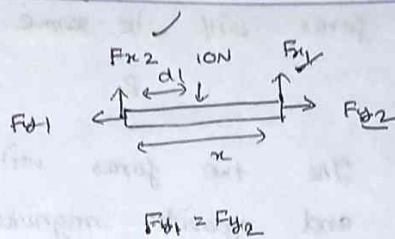


kinetic / static

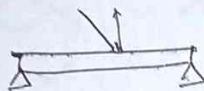
① No of unknown forces = No of equilibrium equation.

② No of unknown forces > No of equilibrium eqns available.

③ Questions



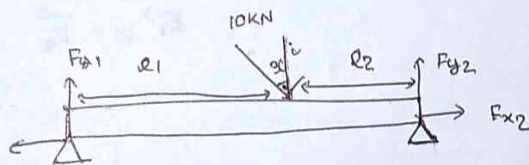
10KN



$$F_{x1} = F_{y2}$$

$$d_1 F_{x2} = x F_{x1} \quad \text{--- ①}$$

→ line of action
→ point of application
→ magnitude

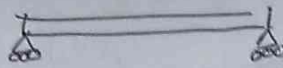


$$F_{x2} \neq 10 \sin 30^\circ \neq F_{x1} \quad \text{--- ①}$$

$$[10 \cos 30^\circ] l_1 = F_{y2} [l_1 + l_2]$$

$$[10 \cos 30^\circ] l_2 = F_{y1} [l_1 + l_2]$$

③ eqns



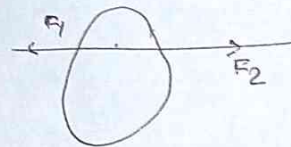
→ ill-posed problem —
 $\frac{\text{no of unknowns} < \text{no of equations}}$

* A body is subjected to two forces, then the equilibrium condition is that

→ The line of action of the two forces will be same.

&

The two forces will have equal and opposite magnitude.



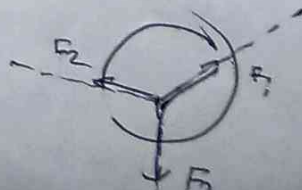
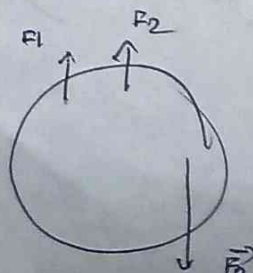
$$\vec{F}_1 = -\vec{F}_2$$

→ If a body is subjected to three forces, then the equilibrium condition will be achieved

① The line of action of three forces will be parallel to each other.

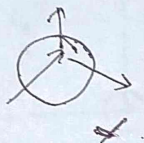
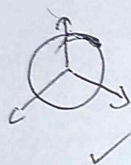
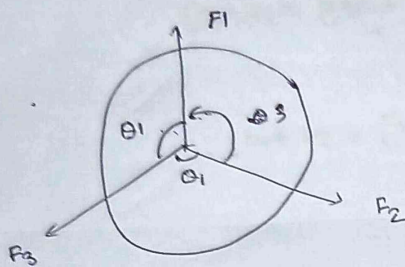
OR

② The lines of action of the three forces will be concurrent in nature.

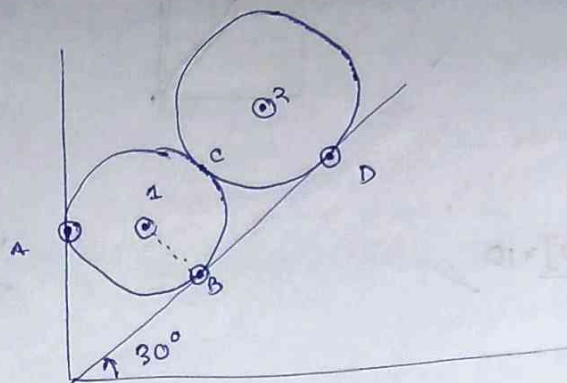


when three coplanar and concurrent forces act on a point under equilibrium, then each force is proportional to sine of angle between other two forces

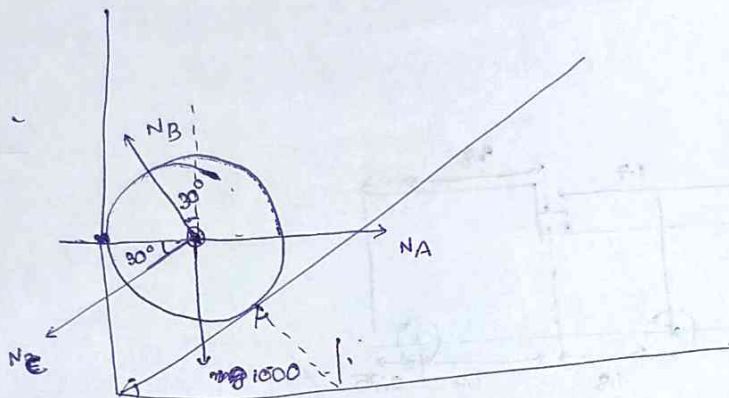
$$\frac{F_1}{\sin \theta_1} = \frac{F_2}{\sin \theta_2} = \frac{F_3}{\sin \theta_3}$$



Q

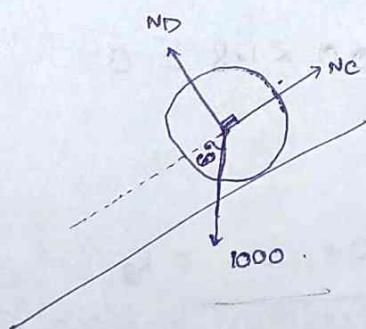


Soluⁿ 0 →



$$N_B \cos 30^\circ = N_C \sin 30^\circ + 1000 \quad \text{--- (1)}$$

$$N_B \sin 30^\circ + N_C \cos 30^\circ = N_A$$



$$\frac{1000}{\sin 30^\circ} \geq \frac{N_C}{\sin(90+60)}$$

$$1000 \times \frac{1}{2} \geq N_C$$

$$N_C \geq 500$$

$$\frac{N_D}{\sin 60^\circ} = \frac{1000}{\sin 30^\circ}$$

$$N_D = 1000 \times \frac{\sqrt{3}}{2}$$

$$N_D = 500\sqrt{3}$$

$$N_B \frac{\sqrt{3}}{2} = N_C \frac{1}{2} + 1000$$

$$N_B \frac{\sqrt{3}}{2} = 250 + 1000$$

$$N_B = \frac{1250 \times 2}{\sqrt{3}}$$

$$N_B = \frac{2500}{\sqrt{3}}$$

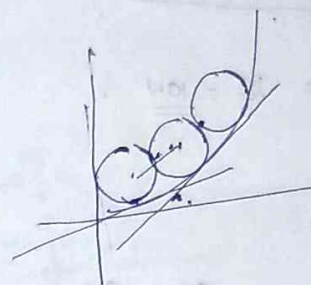
$$\frac{1250}{\sqrt{3}} \times \frac{1}{2} + \frac{250}{500} \times \frac{\sqrt{3}}{2} = N_A$$

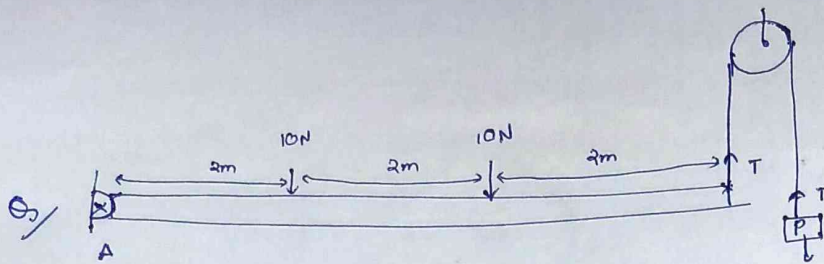
$$\begin{array}{r} 1250 \\ 250 \\ \hline 2000 \end{array}$$

$$\frac{1250\sqrt{3}}{3} + 250\sqrt{3} = N_A$$

$$\Rightarrow N_A = \frac{2000}{\sqrt{3}}$$

$$\frac{2000\sqrt{3}}{3} = N_A$$

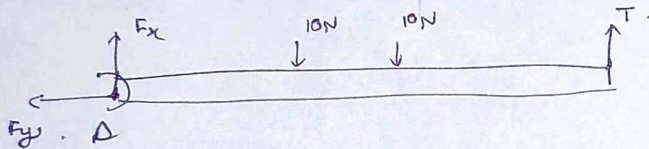




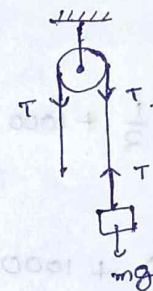
Taking moment about A, we've

$$4 \times 3 = 10 \times 2 + 10 \times 1$$

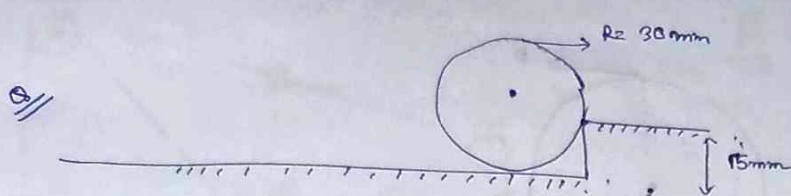
$$T = \frac{30}{3} = \underline{10\text{N}}$$



FBD of beam

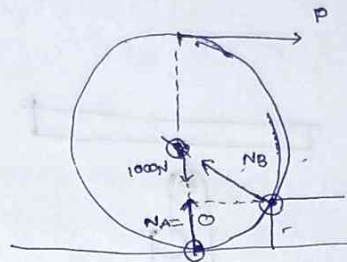


FBD of pulley + block



$$\frac{100}{\sin 30} = \frac{P}{\sin C}$$

Calculate the magnitude of force P req. to start the roller over this curve

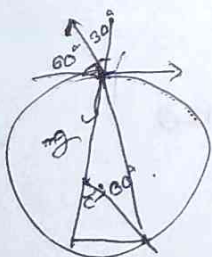


$$\frac{15}{15\sqrt{3}}$$

By moment CORRECT

$$1000 \times 15\sqrt{3} = P \times 15 \times 3$$

$$P = \frac{1000}{\sqrt{3}}$$



$$\sin \theta = \frac{15}{30} = \frac{1}{2}$$

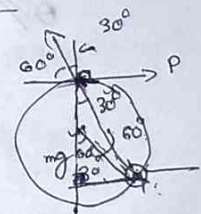
WRONG

$$N \sin 30 = 1000$$

$$N = 2 \times 1000$$

$$N = 2000 \text{ N}$$

$$P = N \cos 30 = 2000 \frac{\sqrt{3}}{2} = 1000\sqrt{3}$$

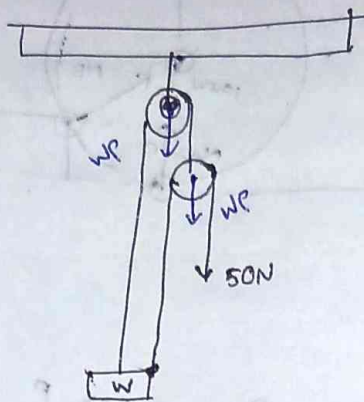


$$\frac{P}{\sin 30} = \frac{1000}{\sin 60}$$

$$P = 1000$$

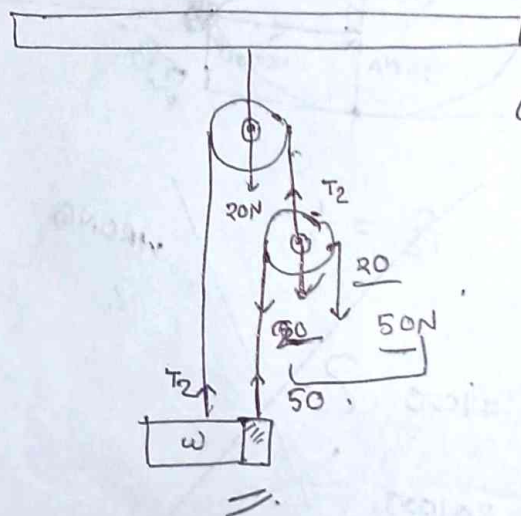
$$\frac{1000}{\sin 30} = \frac{P}{\sin 60} \Rightarrow \frac{1}{\frac{1}{2}} = \frac{P}{\frac{\sqrt{3}}{2}} \Rightarrow P = 1000$$

$$\frac{1000}{\sqrt{3}} = \underline{\underline{P}}$$



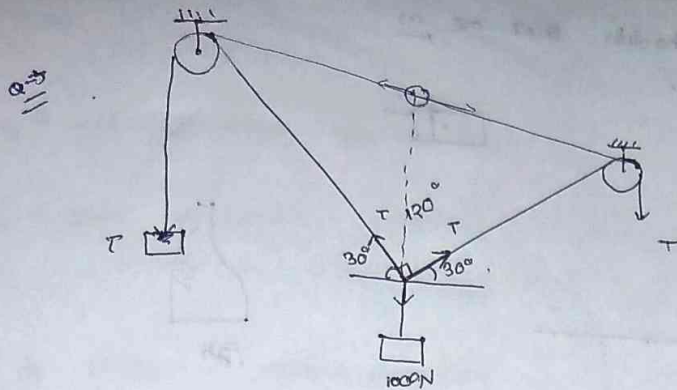
$$w_p = 20$$

Q.4



← its at result
part

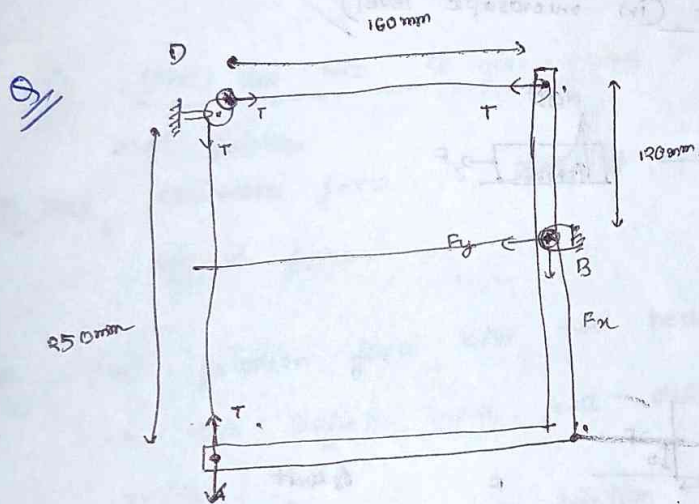
$$50 + 50 + 20 = 120$$



$$\frac{180}{60} = \frac{120}{120}$$

String can be never horizontal

$$\frac{1000}{\sin 120} = T$$



$$F_y \times 120 = 160 F_x$$

$$F_y = \frac{4}{3} F_x$$

$$F_x + 75 = T$$

$$F_y = T$$

$$T = \frac{4}{3} F_x$$

$$\frac{4}{3} T + 75 = T$$

$$\frac{3}{4} T = F_x$$

$$75 = \frac{T}{4}$$

$$T = 75 \times 4$$

$$T = 300$$

$$F_x = \frac{3}{4} \times 300$$

$$F_y = 300$$

$$\sqrt{\frac{300^2 + 28^2}{20}}$$

$$\frac{25 \times 21}{22.5}$$

$$\frac{75 \times 28}{300}$$